Chapter 12: The Cell Cycle & Mitosis

Complete Master Reference Edition • Fused Curriculum (Introduction, DNA & Chromosomes, The Cell Cycle, Lecture Deck Synthesis, Vocabulary & Practice Problems)

Course: BSC2010C General Biology I **Institution:** University of Central Florida **Curriculum Block:** Block 4 (Exam 4

Section 12.1: Content Introduction & DNA Organization

Biological purpose of cell division, evolutionary imperatives, and structural hierarchy from double helix to condensed metaphase chromosome.

Chapter 12 Content Introduction

Learning Objectives

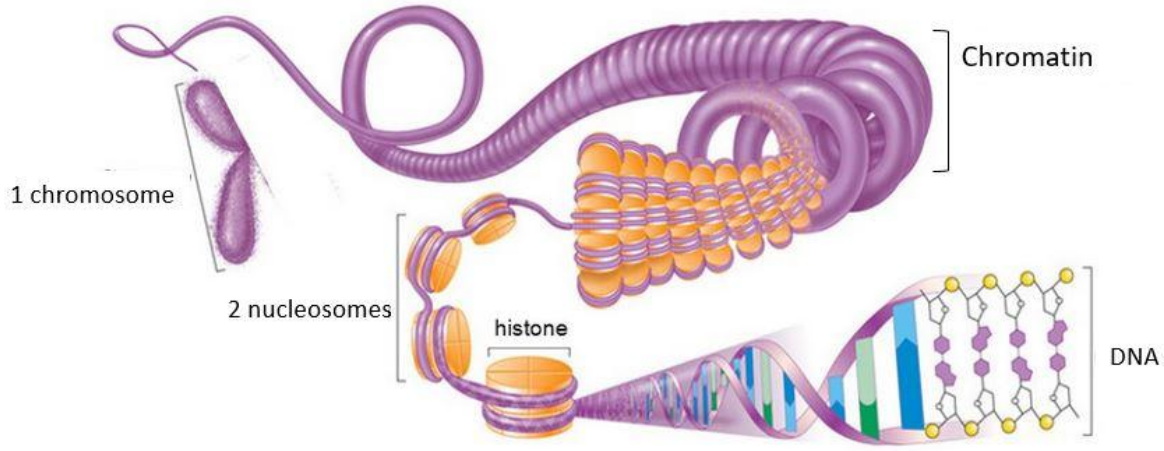
- Explain the different between diploid and haploid
- Distinguish between homologs and sister chromatids
- Explain the difference between unreplicated and replicated chromosomes
- Describe the cell cycle in Eukaryotes
- Describe the events of Interphase
- Describe and recognize the phases of Mitosis
- Explain how checkpoints exert control over the cell cycle

Chapter 12: The Cell Cycle

In Chapter 6, we learned that one tenant of Cell Theory is that cells must come from existing cells. In other words, cells reproduce to create more cells. This ability to reproduce is one thing that separates living organisms from nonliving entities. Different types of cells divide for different reasons. For example, a prokaryotic cell (remember these are single-celled organisms) divides through binary fission to create an exact copy of itself so that the "daughter" is genetically identical to the parent. The same is true of protists such as amoeba. This type of division is called asexual reproduction. The cells in Eukaryotic organisms divide for tissue repair, for growth, and for reproduction. Mitosis is a type of eukaryotic cell division in which there is equal division of genetic material in the nucleus (DNA) and the organelles in the cytoplasm between two cells such that the daughter cell is genetically identical to the parent cell. This sounds a bit like binary fission in prokaryotes but the mechanisms are quite different. Eukaryotic cells use Mitosis to repair tissue (a cut, for example) and for growth (a puppy growing into a full grown dog, for example).

Before we dive into the specifics of Mitosis, we need to understand how DNA is packaged inside cells. Cells contain a very large amount of DNA - almost 2 meters long if you could unwind it! How can 2 meters of DNA fit inside a microscopic cell? Organization! You already know that DNA is double stranded. The double stranded DNA wraps about proteins called **histones** to form a **nucleosome**. The nucleosomes further condense into cylindrical fibers called **chromatin**. Chromatin is further condensed and organized into individual chromosomes.

Each chromosome contains hundreds of genes. Every species has a specific number of chromosomes in the nucleus of their cells: Humans have 46 chromosomes (23 pairs), dogs have 78 chromosomes (39 pairs), etc.



Interactive Course Module: DNA Organization widget

CHAPTER READING

Chapter 12 Textbook Reading: DNA and Chromosomes, The Cell Cycle

CHAPTER LINKS

Chapter 12 PowerPoint: Part 1 ; Part 2

Chapter 12 Post-Quiz

Chapter 12 and 13 Vocabulary

Section 12.2: DNA and Chromosomes

Genome architecture across domains: circular prokaryotic genophores vs. linear eukaryotic chromatin, histones, nucleosomes, euchromatin, and heterochromatin.

Chapter 12 DNA and chromosomes

Learning Objectives

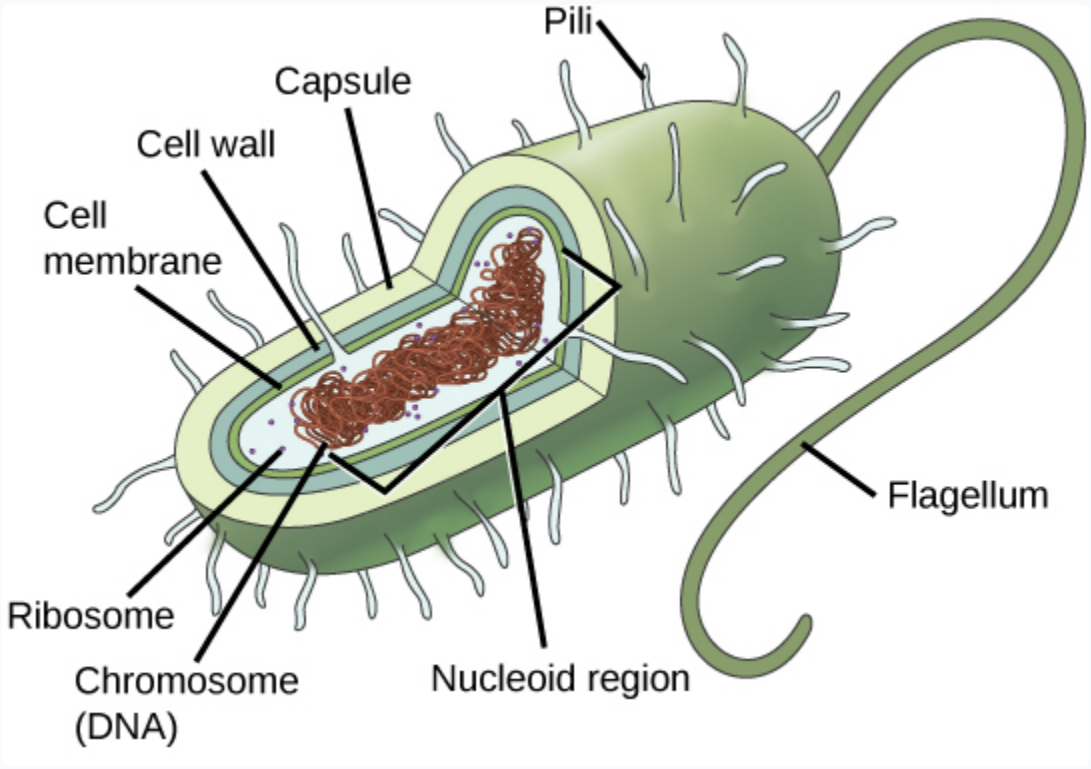
By the end of this section, you will be able to do the following:

- Describe the structure of prokaryotic and eukaryotic genomes
- Distinguish between chromosomes, genes, and traits
- Describe the mechanisms of chromosome compaction

The continuity of life from one cell to another has its foundation in the reproduction of cells by way of the cell cycle. The cell cycle is an orderly sequence of events that describes the stages of a cell's life from the division of a single parent cell to the production of two new genetically identical daughter cells.

Genomic DNA

Before discussing the steps a cell must undertake to replicate and divide its DNA, a deeper understanding of the structure and function of a cell's genetic information is necessary. A cell's DNA, packaged as a double-stranded DNA molecule, is called its genome. In prokaryotes, the genome is composed of a single, double-stranded DNA molecule in the form of a loop or circle (Figure 10.2). Some prokaryotes also have smaller loops of DNA called plasmids that are not essential for normal growth.

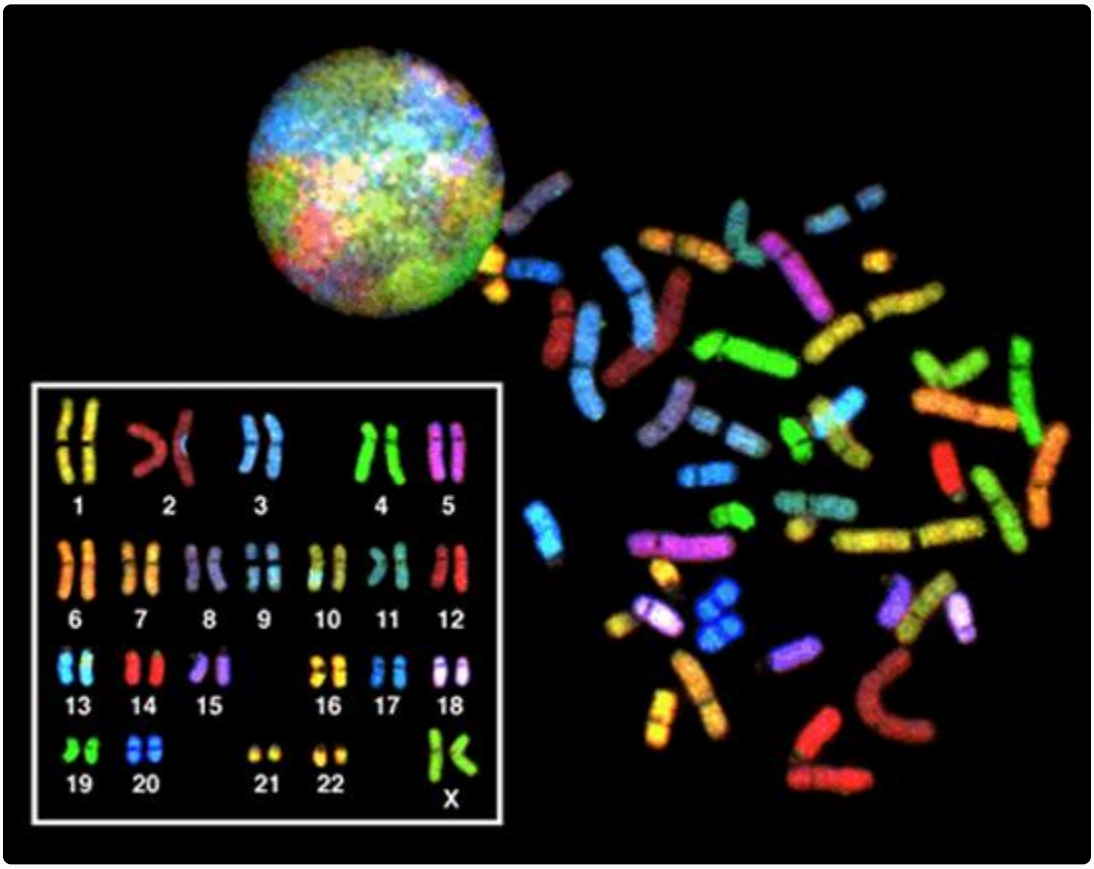


The illustration shows a prokaryotic cell with a single, circular chromosome floating free in the cytoplasm.

Figure 10.2 Prokaryotes, including both Bacteria and Archaea, have a single, circular chromosome located in a central region called the nucleoid.

In eukaryotes, the genome consists of several double-stranded linear DNA molecules called chromosomes (Figure 10.3). Each species of eukaryotes has a characteristic number of chromosomes in the nuclei of its cells. Human body (somatic) cells have 46 chromosomes, while human gametes (sperm or eggs) have 23 chromosomes each. A typical body cell contains two matched or homologous sets of chromosomes (one set from each biological parent)—a configuration known as diploid. (Note: The letter *n* is used to represent a single set of chromosomes; therefore, a diploid organism is designated 2*n*.) Human cells that contain one set of chromosomes are called gametes, or sex cells; these are eggs and sperm, and are designated *1n*, or haploid.

Upon fertilization, each gamete contributes one set of chromosomes, creating a diploid cell containing matched pairs of chromosomes called homologous ("same knowledge") chromosomes. Homologous chromosomes are the same length and have specific nucleotide segments called genes in exactly the same location, or locus. Genes, the functional units of chromosomes, determine specific characteristics by coding for specific proteins. Traits are the variations of those characteristics. For example, hair color is a characteristic with traits that are brown, black, blonde and many colors in between.



The 23 chromosomes from a human female are each dyed a different color so they can be distinguished. During most of the cell cycle, each chromosome is elongated into a thin strand that folds over on itself, like a piece of spaghetti. The ch...

Figure 10.3 There are 23 pairs of homologous chromosomes in a female human somatic cell. The condensed chromosomes are viewed within the nucleus (top), removed from a cell during mitosis (also called karyokinesis or nuclear division) and spread out on a slide (right), and artificially arranged according to length (left); an arrangement like this is called a karyotype. In this image, the chromosomes were exposed to fluorescent stains for differentiation of the different chromosomes. A method of staining called "chromosome painting" employs fluorescent dyes that highlight chromosomes in different colors. (credit: National Human Genome Project/NIH)

Each copy of a homologous pair of chromosomes originates from a different parent; therefore, the different genes (alleles) themselves are not identical, although they code for the same traits such as "hair color." The variation of individuals within a species is due to the specific combination of the genes inherited from both parents. The sex chromosomes, X and Y, are the single exception to the rule of homologous chromosome uniformity: Other than a small amount of homology that is necessary to accurately produce gametes, the genes found on the X and Y chromosomes are different.

Eukaryotic Chromosomal Structure and Compaction

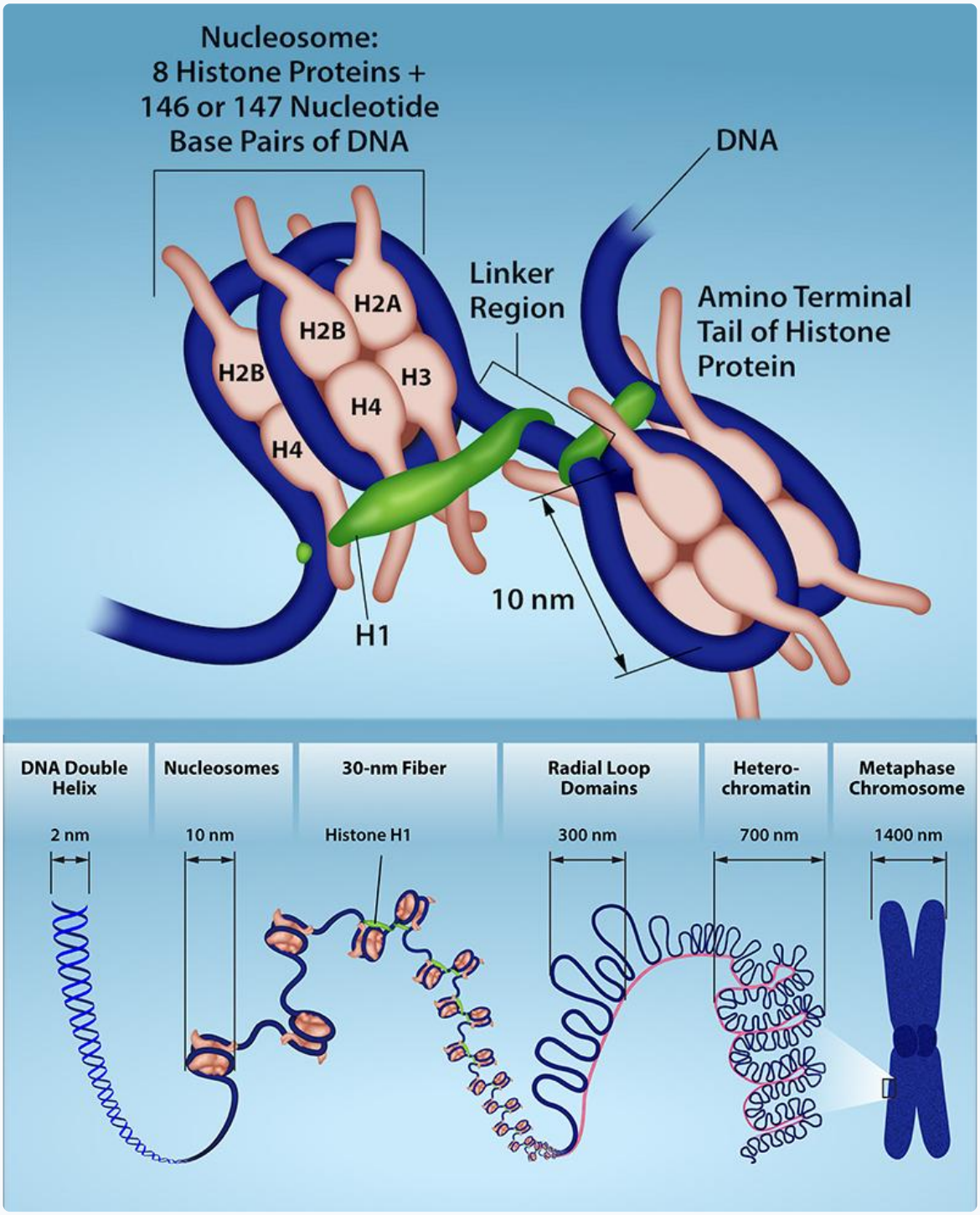
If the DNA from all 46 chromosomes in a human cell nucleus were laid out end-to-end, it would measure approximately two meters; however, its diameter would be only 2 nm! Considering that the size of a typical human cell is about 10 µm (100,000 cells lined up to equal one meter), DNA must be *tightly packaged* to fit in the cell's nucleus. At the same time, it must also be readily accessible for the genes to be expressed. There are a number of ways that chromosomes are compacted.

In the first level of compaction, short stretches of the DNA double helix wrap around a core of eight histone proteins at regular intervals along the entire length of the chromosome (Figure 10.4). The DNA-histone complex

is called **chromatin**. The beadlike, histone DNA complex is called a nucleosome. A DNA molecule in this form is about seven times shorter than the double helix without the histones.

The second level of compaction occurs as the nucleosomes coil into a chromatin fiber. This coiling further *condenses* the chromosome so that it is now about 50 times shorter than the extended form.

In the third level of compaction, a variety of *fibrous proteins* is used to "pack the chromatin." These fibrous proteins also ensure that each chromosome in a non-dividing cell occupies a particular area of the nucleus that does not overlap with that of any other chromosome (see the top image in Figure 10.3).



A detail of a nucleosme shows 8 Histone proteins and 146 or 147 Nucleotide Base pairs; they are connected by a linker region to the Amino tail of the histone protein. Then, several levels of magnification reveal details of DNA. A D N A ...

Figure 10.4 Each linear chromosome in a eukaryotic cell is packaged into chromatin, a combination of DNA and proteins. The double-stranded DNA helix associates with the core histones to form nucleosomes. These nucleosomes are further organized into a 30 nm fiber by the linker histone, H1. The fiber then associates with additional proteins to form loops and higher-order heterochromatin packing. DNA packing reaches its most condensed state during metaphase in mitosis in preparation for chromosome separation. Chromatin packing is

dynamic and undergoes reversible changes in response to changes in gene expression and the cell cycle. Credit: Rao, A., Ryan, K. Fletcher, S. Hawkins, A. and Tag, A. Department of Biology, Texas A&M University.

Section 12.3: The Cell Cycle, Mitosis & Regulatory Checkpoints

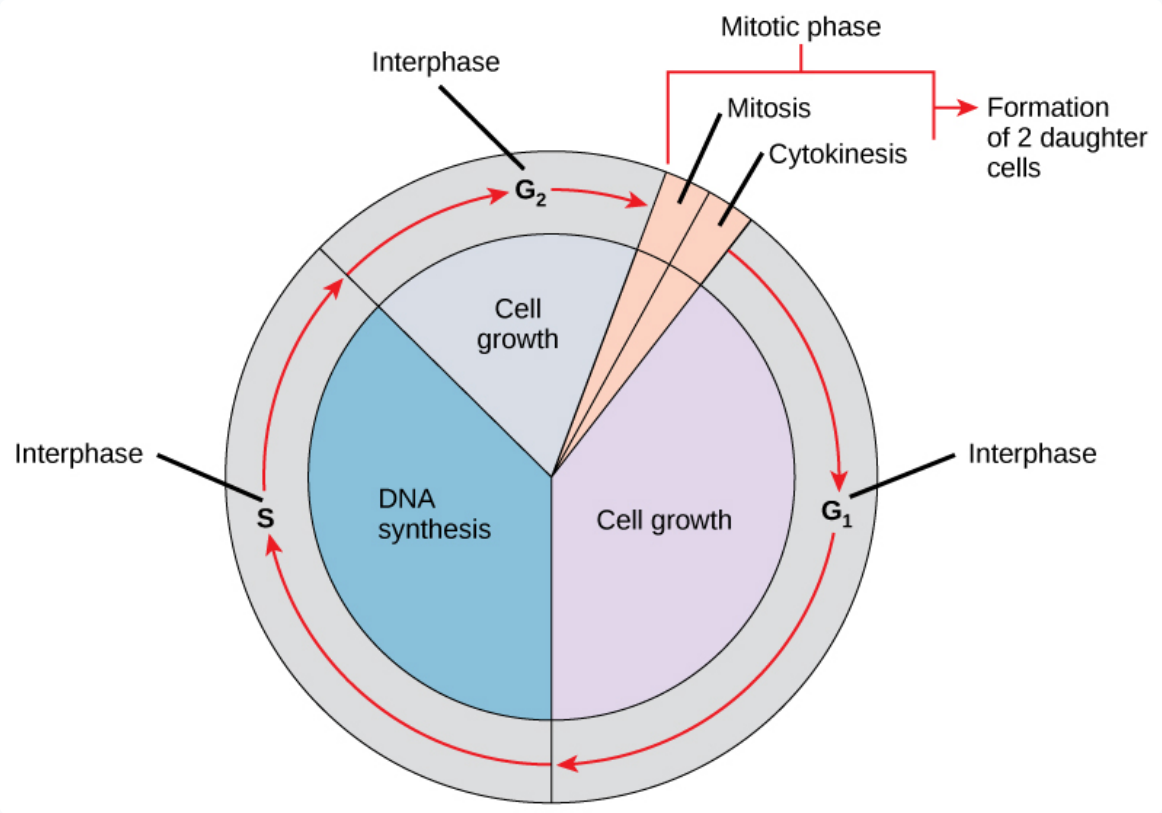
Interphase stages (G1, S, G2), five mitotic phases (Prophase, Prometaphase, Metaphase, Anaphase, Telophase), cytokinesis mechanisms, and molecular checkpoints (G1, G2/MPF, M/Spindle).

Chapter 12 The Cell Cycle

Learning Objectives

- Describe the three stages of interphase
- Discuss the behavior of chromosomes during mitosis
- Explain how the cytoplasmic content is divided during cytokinesis
- Define the G_0 phase

The cell cycle is an ordered series of events involving cell growth and cell division that produces two new daughter cells. Cells on the path to cell division proceed through a series of precisely timed and carefully regulated stages of growth, DNA replication, and nuclear and cytoplasmic division that ultimately produces two identical cells. The cell cycle has two major phases: interphase and the mitotic phase (Figure 10.5). During interphase, the cell grows and DNA is replicated. During the mitotic phase, the replicated DNA and cytoplasmic contents are separated, and the cell cytoplasm is typically partitioned by a third process of the cell cycle called cytokinesis.



Like a clock, the cell cycles from interphase to the mitotic phase and back to interphase. Most of the cell cycle is spent in interphase, which is subdivided into G subscript 1 baseline, S, and G subscript 2 baseline phases. Cell growth occ...

Figure 10.5 The cell cycle in multicellular organisms consists of interphase and the mitotic phase. During interphase, the cell grows and the nuclear DNA is duplicated. Interphase is followed by the mitotic phase. During the mitotic phase, the duplicated chromosomes are segregated and distributed into daughter nuclei. Following mitosis, the cytoplasm is usually divided as well by cytokinesis, resulting in two genetically identical daughter cells.

Interphase

During interphase, the cell undergoes normal growth processes while also preparing for cell division. In order for a cell to move from interphase into the mitotic phase, many internal and external conditions must be met. The three stages of interphase are called *G_1, S, and G_2*.

G_1 Phase (First Gap)

The first stage of interphase is called the G_1 phase (first gap) because, from a microscopic point of view, little change is visible. However, during the G_1 stage, the cell is quite active at the biochemical level. The cell is accumulating the building blocks of chromosomal DNA and the associated proteins as well as accumulating sufficient energy reserves to complete the task of replicating each chromosome in the nucleus.

S Phase (Synthesis of DNA)

Throughout interphase, nuclear DNA remains in a semi-condensed chromatin configuration. In the S phase, DNA replication can proceed through the mechanisms that result in the formation of identical pairs of DNA molecules —sister chromatids—that are firmly attached at the centromere. The centrosome is also duplicated during the S

phase. The two centrosomes will give rise to the mitotic spindle, the apparatus that orchestrates the movement of chromosomes during mitosis.

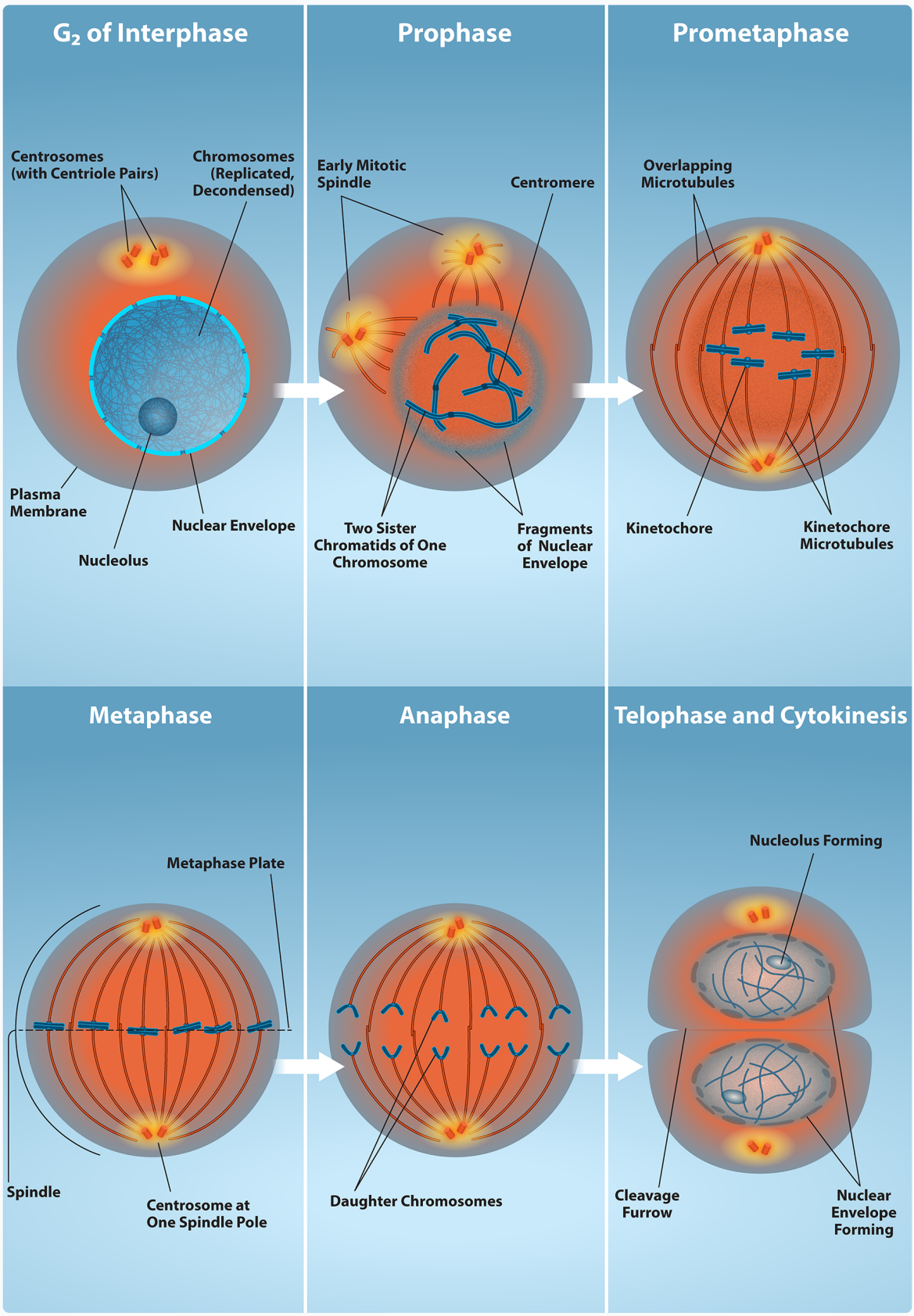
G_2 Phase (Second Gap)

In the G_2 phase, the cell replenishes its energy stores and synthesizes proteins necessary for chromosome manipulation and movement. Some cell organelles are duplicated, and the cytoskeleton is dismantled to provide resources for the mitotic phase. There is additional cell growth during G_2. The final preparations for the mitotic phase must be completed before the cell is able to enter the first stage of mitosis.

The Mitotic Phase

The mitotic phase is a multistep process during which the duplicated chromosomes are aligned, separated, and move into two new, identical daughter cells. The first portion of the mitotic phase is nuclear division. As we have just seen, the second portion of the mitotic phase is called cytokinesis—the physical separation of the cytoplasmic components into the two daughter cells.

VISUAL CONNECTION

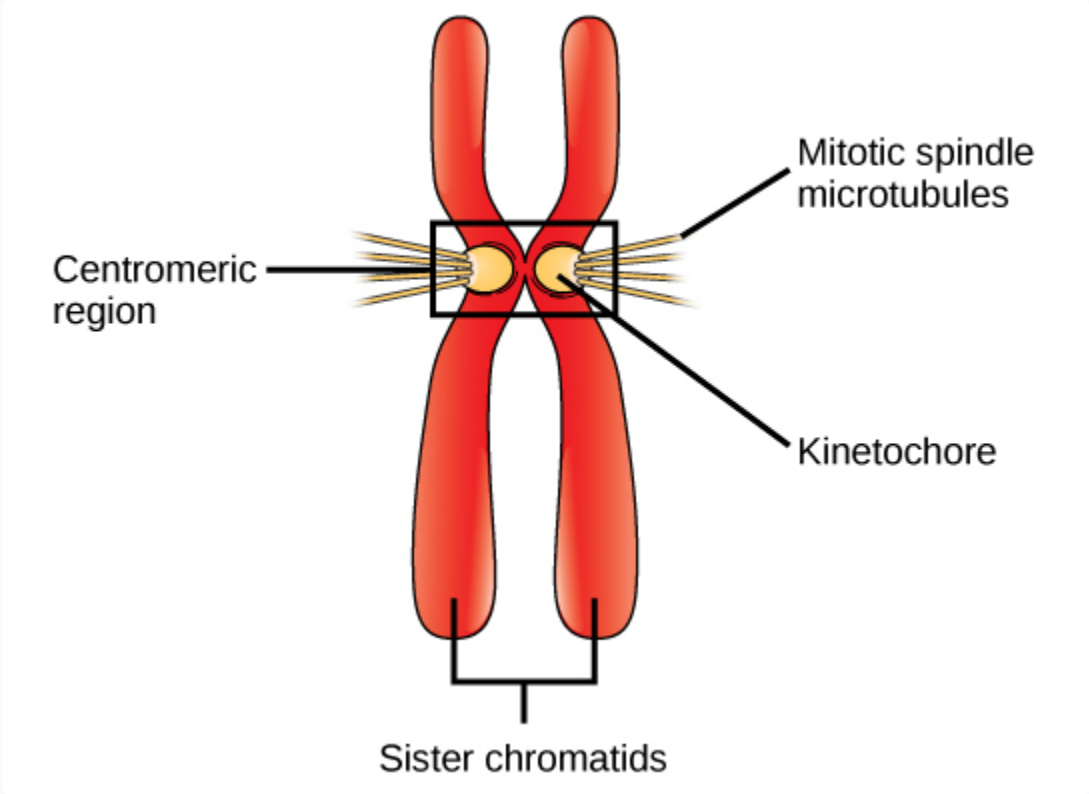


This diagram shows the five phases of mitosis and cytokinesis. During prophase, the chromosomes condense and become visible, spindle fibers emerge from the centrosomes, the nuclear envelope breaks down, and the nucleolus disappears. During ...

Figure 10.6 G2 of Interphase – The last stage of interphase is the second gap period, G2. During this stage, cells grow, replenish energy and synthesize needed macromolecules, such as proteins and lipids.
Mitosis - When G2 is complete, the cell will enter mitosis. Although there are 5 phases in mitosis, with the exception of the metaphase to anaphase transition, these phases are not discrete and happen as a continuous process. Prophase is the first stage in mitosis. The nuclear envelope begins to break down and chromosomes condense and are now visible. Spindle fibers start to appear and centrosomes begin to move towards opposite poles.
Prometaphase - Chromosomes continue to condense and are more visible. Kinetochores appear at the centromere and kinetochore microtubules attach. Centrosomes continue to move towards opposite poles.
Metaphase – The mitotic spindle is fully developed and centrosomes are at opposite poles. Chromosomes are aligned at the "equatorial plate", and each sister chromatid rests on one side of the plate, with spindle fibers attached to them.
Anaphase - Sister chromatids are pulled apart by spindle fibers and are separated from each other. Each chromatid is now a chromosome.
Telophase and Cytokinesis - Chromosomes arrive at opposite poles and start to decondense and become less visible. The nuclear envelope reassembles and begins to surround each new set of chromosomes. The mitotic spindle assembly breaks down and the division of the cytoplasm begins via cytokinesis. This physical separation into two cells are remarkably different processes in plant and animal cells.
Credit: Rao, A., Hawkins, A.and Fletcher, S. Department of Biology, Texas A&M University.

Prophase: the nuclear envelope starts to dissociate into small vesicles, and the membranous organelles (such as the Golgi complex [Golgi apparatus] and the endoplasmic reticulum), fragment and disperse toward the periphery of the cell. The nucleolus disappears (disperses) as well, and the centrosomes begin to move to opposite poles of the cell. Microtubules that will form the *mitotic spindle* extend between the centrosomes, *pushing them farther apart* as the microtubule fibers lengthen. The sister chromatids begin to coil more tightly with the aid of condensin proteins and now become visible under a light microscope.

Prometaphase: Many processes that began in prophase continue to advance. The remnants of the nuclear envelope fragment further, and the mitotic spindle continues to develop as more microtubules assemble and stretch across the length of the former nuclear area. Chromosomes become even more condensed and discrete. Each sister chromatid develops a protein structure called a kinetochore in its centromeric region (Figure 10.7). The proteins of the kinetochore attract and bind to the mitotic spindle microtubules. As the spindle microtubules extend from the centrosomes, some of these microtubules come into contact with and firmly bind to the kinetochores. Once a mitotic fiber attaches to a chromosome, the chromosome will be oriented until the kinetochores of sister chromatids face the *opposite poles*. Eventually, all the sister chromatids will be attached via their kinetochores to microtubules from opposing poles.



This illustration shows two sister chromatids. Each has a kinetochore at the centromere, and mitotic spindle microtubules radiate from the kinetochore.

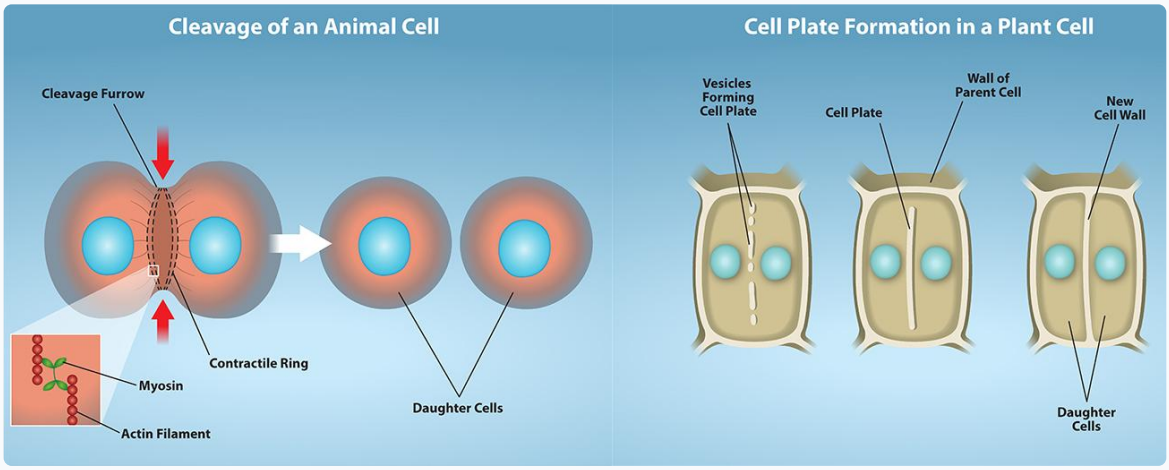
Figure 10.7 During prometaphase, mitotic spindle microtubules from opposite poles attach to each sister chromatid at the kinetochore. In anaphase, the connection between the sister chromatids breaks down, and the microtubules pull the chromosomes toward opposite poles.

Metaphase: All the chromosomes are aligned in a plane called the metaphase plate, or the equatorial plane, roughly midway between the two poles of the cell. The sister chromatids are still tightly attached to each other at the centromere.

Anaphase: The cohesin proteins degrade, and the sister chromatids separate at the centromere. Each chromatid, now called a daughter chromosome, is pulled rapidly toward the centrosome to which its microtubule is attached. The cell becomes visibly elongated (oval shaped) as the polar microtubules slide against each other at the metaphase plate where they overlap.

Telophase: the chromosomes reach the opposite poles and begin to *decondense* (unravel), relaxing once again into a stretched-out chromatin configuration. The mitotic spindles are depolymerized into tubulin monomers that will be used to assemble cytoskeletal components for each daughter cell. Nuclear envelopes form around the chromosomes, and nucleosomes appear within the nuclear area.

Cytokinesis: a contractile ring composed of actin filaments forms just inside the plasma membrane at the former metaphase plate. The actin filaments pull the equator of the cell inward, forming a fissure. This fissure is called the cleavage furrow. The furrow deepens as the actin ring contracts, and eventually the membrane is cleaved in two (Figure 10.9).



Part a: This illustration shows cytokinesis in a typical animal cell. Part b: Cytokinesis is shown in a typical plant cell. In an animal cell, a contractile ring of actin filaments forms a cleavage furrow that divides the cell in two. In a ...

Figure 10.9 Animal cell cleavage - Contraction of actin filaments by myosin pull the "equator" of the cell causing an invagination of the cell (a fissure called a cleavage furrow). The cleavage furrow gets continuously deeper until it eventually divides the cell into two new, independent daughter cells.
Plant cell wall formation - A mixture of enzymes, proteins, and glucose molecules are transported via vesicles to the center of the cell. These vesicles continuously build upon each other until a completely new cell wall has emerged and two new cells are formed, independent of one another.
Credit: Rao, A., Ryan, K., Hawkins, A.and Fletcher, S. Department of Biology, Texas A&M University.

G_0 Phase

Cells in G_0 phase are not actively preparing to divide. The cell is in a stage that occurs when cells exit the cell cycle. Some cells enter G_0 temporarily due to environmental conditions such as availability of nutrients, or stimulation by growth factors. The cell will remain in this phase until conditions improve or until an external signal triggers the onset of G_1. Other cells that never or rarely divide, such as mature cardiac muscle and nerve cells, remain in G_0 permanently.

Section 12.4: Instructor Lecture Slide Review (51 Slides Synthesis)

Comprehensive pedagogical synthesis of Dr. Dawson's lecture slide presentations: Part 1 (23 slides) and Part 2 (28 slides).

Lecture Part 1 (Slides 1–23): DNA Organization, Chromosome Biology & Karyotypes

- **Genetic Units:** DNA carries hereditary genes organized into chromosomes. Prokaryotes possess a single, circular chromosome in a nucleoid region with minimal associated proteins. Eukaryotes maintain multiple, linear chromosomes packaged into chromatin.
- **Anatomy of a Duplicated Eukaryotic Chromosome:**
  - **Centromere:** Narrow "waist" region of repetitive DNA where sister chromatids are most closely attached via cohesin proteins; assembly site for the proteinaceous **kinetochore**.
  - **Chromosome Arms:** Segments extending above (p-arm) and below (q-arm) the centromere.
  - **Telomeres:** Specialized non-coding repetitive nucleotide sequences capping the physical ends of linear chromosomes, protecting against degradation and end-joining.
- **Karyotypic Invariance & Species Chromosome Numbers:** Each species maintains a constant diploid chromosome number (e.g., Humans: 2n = 46; Dogs: 2n = 78; Chimpanzees: 2n = 48; Horses: 2n = 64; Grevy's Zebra: 2n = 46). Chromosome number correlates neither with organismal complexity nor intelligence, but closely related taxa display conserved karyotypes.
- **Homologous Chromosomes:** Somatic cells are diploid (2n), containing two matched chromosomes for each karyotypic pair: one paternal homolog and one maternal homolog. Homologs share the same length, centromere position, banding pattern, and gene loci, but frequently carry distinct versions of those genes (alleles).
- **Chromatin Compaction Hierarchy:**
  1. DNA Double Helix (2 nm width).
  2. **Nucleosomes ("Beads on a String", 10 nm):** DNA wound twice around an octamer core of 8 basic histone proteins (two each of H2A, H2B, H3, H4) encompassing ~146-147 base pairs; linker DNA bound by histone H1 connects adjacent beads.
  3. 30-nm Chromatin Fiber: Helical coiling of nucleosomes.
  4. 300-nm Looped Domains: Fibers attached to a non-histone protein scaffold.
  5. Condensed Metaphase Chromosome (1400 nm): Maximal compaction ensuring safe, untangled segregation during anaphase.

Lecture Part 2 (Slides 1–28): The Cell Cycle Clock, Mitosis & Molecular Controls

- **Cell Cycle Architecture:** Composed of Interphase (~90% of cycle duration) and the Mitotic (M) Phase.
  - **G1 Phase (First Gap):** Cell growth, high metabolic activity, organelle duplication, accumulation of protein building blocks and energy.
  - **S Phase (DNA Synthesis):** Precise semiconservative replication of the entire nuclear genome. Chromosomes duplicate from single chromatids to paired sister chromatids joined at the centromere. Centrosomes duplicate.
  - **G2 Phase (Second Gap):** Further cell growth, synthesis of tubulin and mitotic proteins, preparation for spindle assembly, final checkpoint checks.
- **The Mitotic Spindle & Kinetochores:**
  - **Centrosome:** The microtubule-organizing center (MTOC) containing a pair of centrioles in animal cells. Replicated in interphase; moves to opposite poles during prophase/prometaphase.
  - **Kinetochores:** Multi-protein assemblies formed on centromeric DNA (one per sister chromatid, facing opposite directions). Kinetochore microtubules capture chromosomes and exert pulling tension. Non-kinetochore microtubules overlap at the equator and elongate the cell.
- **The Five Phases of Mitosis:**
  1. **Prophase:** Chromatin condenses into visible discrete chromosomes; nucleoli disappear; mitotic spindle begins forming; centrosomes push apart.
  2. **Prometaphase:** Nuclear envelope fragments; spindle microtubules invade the nuclear region; kinetochores mature on centromeres and are captured by microtubules.
  3. **Metaphase:** Centrosomes reach opposite spindle poles; chromosomes convene at the **metaphase plate** (equator), equidistant between poles; kinetochores of sister chromatids are attached to opposite poles under tension.
  4. **Anaphase:** Cohesin proteins are cleaved by the enzyme separase; sister chromatids instantaneously separate into independent individual chromosomes; motor proteins walk chromosomes poleward along depolymerizing microtubules; non-kinetochore microtubules push poles apart.
  5. **Telophase:** Two daughter nuclei form; nuclear envelopes assemble from fragments; nucleoli reappear; chromosomes decondense back into diffuse chromatin; spindle depolymerizes.
- **Cytokinesis Mechanisms:**
  - **Animal Cells:** Formation of a **cleavage furrow**; a contractile ring of actin microfilaments and myosin motor proteins pinches the plasma membrane inward like a drawstring until two separate cells pinch off.
  - **Plant Cells:** Cell walls prevent furrowing. Golgi-derived vesicles carrying cell wall materials coalesce along the metaphase plate, forming a **cell plate** that expands outward until fusing with the parental cell wall.
- **Molecular Regulation & Internal Checkpoints:**
  - **G1 Checkpoint ("The Restriction Point"):** The critical master checkpoint. Assesses cell size, nutrient reserves, growth factor signals, and DNA integrity. If the "go-ahead" signal is received, the cell completes S, G2, and M. In the absence of go-ahead, the cell exits the cycle into **G0 phase**, a non-dividing quiescent state (e.g., mature neurons, cardiac myocytes).
  - **G2 Checkpoint & MPF (Maturation / Mitosis-Promoting Factor):** Checks for completed DNA replication and DNA damage. Formed by the union of a **Cyclin-Dependent Kinase (Cdk)** and regulatory **cyclin**. Cyclin accumulates during late S and G2; active MPF phosphorylates nuclear

lamins (triggering envelope breakdown) and initiates mitosis. MPF activity peaks at metaphase; during anaphase, cyclin is degraded, inactivating Cdk and permitting mitotic exit.

- **M Checkpoint (Spindle Assembly Checkpoint):** Occurs during metaphase. Ensures that every single kinetochore is firmly attached to spindle fibers under bilateral tension. Unattached kinetochores emit a molecular stop signal, delaying anaphase until proper attachment is confirmed, preventing nondisjunction.

- **External Growth Controls & Cancer Pathophysiology:**
  - **Density-Dependent Inhibition:** Crowded cells stop dividing when cell-surface proteins contact neighbors.
  - **Anchorage Dependence:** Most animal cells must be anchored to a physical substratum (extracellular matrix or tissue culture dish) to divide.
  - **Cancer Cells:** Cancer cells ignore cell cycle control signals, bypass checkpoints, synthesize their own growth factors or possess constitutively active signal transduction pathways, evade apoptosis, lose density-dependent inhibition (forming tumors), and lose anchorage dependence (allowing local invasion and metastatic colonization).

Section 12.5: Chapter 12 Official Vocabulary & Conceptual Foundations

Key terms transcribed from the official course glossary with precise biological definitions.

Somatic Cell
All body cells in a multicellular organism except sperm and eggs. Somatic cells are diploid (2n). Examples include skin, liver, muscle, and neuron cells.

Gametes
Reproductive haploid cells (sperm and eggs, designated n) that fuse during fertilization to form a diploid zygote.

Autosomes
All chromosomes in a genome that do not determine an individual's biological sex (in humans, chromosome pairs 1 through 22).

Sex Chromosomes
Chromosomes that determine biological sex (X and Y in humans; female = XX, male = XY).

Homologous Chromosomes
A pair of chromosomes of identical length, centromere position, and banding pattern that carry the same gene characters (one inherited from mom, one from dad).

Diploid (2n)
Having two complete sets of chromosomes (one maternal, one paternal). Humans: 2n = 46. Dogs: 2n = 78.

Haploid (n)
Having only a single set of unpaired chromosomes. Found in gametes. Humans: n = 23. Dogs: n = 39.

Sister Chromatids
Two identical copies of a single duplicated chromosome produced during S phase, joined together at the centromere by cohesins.

Centromere
The specialized constricted region of a duplicated chromosome where sister chromatids are bound and where kinetochores assemble.

Kinetochore
A multi-protein disc assembled on centromeric DNA that attaches chromosomes to the spindle microtubules during mitosis and meiosis.

Centrosome
The main microtubule-organizing center (MTOC) of animal cells containing two centrioles; coordinates spindle assembly.

Metaphase Plate
The imaginary equatorial plane through the center of the cell where chromosomes align during metaphase under equal spindle tension.

Section 12.6: Official Course Practice Problems & Verified Solutions

Official examination practice problems from Chapters 12 and 13 Practice Questions.docx with complete answers and explanations.

1. An organism's somatic cells have 24 chromosomes. Which of the following statements is TRUE?

A) n = 24 • **B) Gametes for this organism have 12 chromosomes (Correct)** • C) 2n = 12 • D) There are 24 pairs of homologous chromosomes

Rationale: Somatic cells are diploid (2n = 24), which means n = 12. Gametes are haploid (n = 12). There are 12 homologous pairs, not 24.

2. If a cell does not pass the G1 checkpoint:

A) It will continue on to the S phase • B) It will be "stuck" in G2 phase • **C) It will enter G0 phase (Correct)** • D) It will go straight into mitosis

Rationale: In the absence of a go-ahead signal at the G1 checkpoint (restriction point), the cell switches into the non-dividing G0 state.

3. What is the name of the microtubule-organizing center found in animal cells as an identifiable structure present during all phases of the cell cycle?

A) Centriole • **B) Centrosome (Correct)** • C) Centromere • D) Kinetochore

Rationale: The centrosome is the organelle functioning as the primary MTOC; each centrosome contains a pair of centrioles.

4. Microtubules assist in what process in mitosis and meiosis?

A) Movement of organelles • B) Growth of the cell in G1 and G2 • **C) Movement of chromosomes (Correct)** • D) Creating sister chromatids

Rationale: Kinetochore microtubules pull sister chromatids to opposite poles, while polar microtubules elongate the spindle axis.

5. Metaphase of mitosis is characterized by:

A) Alignment of sister chromatids on the metaphase plate (Correct) • B) Separation of the centromeres • C) Cytokinesis • D) Separation of sister chromatids

Rationale: During mitotic metaphase, individual duplicated chromosomes align single-file along the equatorial metaphase plate.

6. In a diploid cell with 2n = 10, how many sister chromatids will be found in a nucleus at prophase of mitosis?

A) 5 • B) 10 • **C) 20 (Correct)** • D) 40

Rationale: Each of the 10 chromosomes has duplicated during S phase, consisting of 2 sister chromatids: 10 x 2 = 20 sister chromatids.

7. If a cell at metaphase of mitosis contains 80 sister chromatids, how many chromosomes will be present in a G1 cell?

A) 20 • **B) 40 (Correct)** • C) 80 • D) 160

Rationale: 80 sister chromatids represent 40 duplicated chromosomes (80 / 2 = 40). In G1 (prior to S phase replication), there are 40 un-duplicated chromosomes.

8. Which of the following molecules triggers the cell's passage past the G2 checkpoint into mitosis?

A) PDGF • **B) MPF (Maturation / Mitosis-Promoting Factor) (Correct)** • C) Cyclin alone • D) Cdk alone

Rationale: MPF is the active complex of Cdk and cyclin that phosphorylates key mitotic targets to trigger nuclear envelope breakdown and chromatin condensation.

9. Which of the following processes is associated with asexual reproduction?

A) Meiosis • **B) Mitosis (Correct)** • C) Fertilization • D) Exchange of chromosomes between species

Rationale: Mitotic division produces two daughter cells genetically identical to the parent cell (clones), which underlies asexual reproduction in single-celled eukaryotes and vegetative propagation.

10. A cell has 2n = 18. How many sister chromatids are present after S phase?

A) 9 • B) 18 • **C) 36 (Correct)** • D) 72

Rationale: In S phase, each of the 18 chromosomes is replicated into 2 sister chromatids (18 x 2 = 36 chromatids).

11. A cell has 2n = 40. How many chromosomes are present in the cell at G0?

A) 10 • B) 20 • **C) 40 (Correct)** • D) 80

Rationale: Cells in G0 maintain their normal diploid chromosome count (2n = 40) in an unreplicated state.

Appendix: Exam 4 Focus Specifications & Mitosis Ploidy Matrix

Official curriculum requirements from Exam 4 Study Guide 2026.docx covering chromosome dynamics and cell cycle regulation.

Summary Matrix: Chromosome & Chromatid Accounting Through Mitosis (Example: 2n = 46)

Stage	Ploidy Level	Chromosome Number	Sister Chromatid Count	Physical Configuration
G1 Phase	Diploid (2n)	46	0	Unreplicated diffuse chromatin fibers
S Phase (Post)	Diploid (2n)	46	92	Replicated; each chromosome has 2 sister chromatids
G2 Phase	Diploid (2n)	46	92	Double centrosome; chromatin condenses
Prophase / Metaphase	Diploid (2n)	46	92	46 chromosomes aligned single-file at equator
Anaphase	Tetraploid transiently (4n)	**92**	**0**	Cohesins cleaved; chromatids become individual chromosomes!
Telophase / Cytokinesis	Diploid (2n) in each	46 per cell	0	2 identical daughter cells, each with 46 chromosomes

Exam 4 High-Yield Pitfalls & Diagnostic Takeaways for Chapter 12

- **Sister Chromatids vs. Chromosomes in Anaphase:** As soon as sister chromatids separate at the centromere in anaphase, each is legally considered an independent, individual chromosome. The number of chromosomes temporarily doubles from 46 to 92, while the number of chromatids drops to 0!
- **MPF Inactivation:** MPF turns itself off by activating an enzyme complex that degrades its own cyclin subunit during anaphase. Cdk is not destroyed; it remains in the cell awaiting new cyclin in the next cycle.
- **Cytokinesis Timing:** Cytokinesis typically begins in late anaphase or telophase and continues after mitosis completes, partitioning the cytoplasm.